

Contemporary Issues in Computer Science and Engineering

CSE 3000 (Fall 2024)

Instructor: Rye Howard-Stone (Rye.Howard-Stone@UConn.edu)

Office Hours: By appointment

Course Description:

As budding experts in your field of choice, you will soon be presented with situations that require an eye trained towards ethical considerations. Two questions I'd like you to keep in mind, from this semester on, are:

- What are citizens of a community owed by that community?
- What do citizens of a community owe to each other?

You may be in a position to shape the development of technology, progress, and innovation we all experience as a community - sooner than you think. This course explores how these two questions relate to your future careers, with special consideration given to the role of "expert citizens" such as business leaders and engineers when it comes to the institutions they work with and the technology they produce.

Please complete the readings PRIOR to attending class. Participation is a substantial portion of your grade and aims to measure both comprehension of the readings, and your willingness to engage during class.

You must also complete two reflections during the semester (600-800 words each). **The reflections may be completed at any time** after the first discussion class – hand these in well ahead of the due date – **do not turn them in late**. Late submissions are penalized at 15 points per day late.

The final is a group project and presentation that combines the expertise you have in Computer Science with the ethical considerations we explore in this class.

All readings will be provided on HuskyCT

Learning Outcomes:

At the end of this course, you will be able to:

- Identify ethical arguments in a wide variety of settings
- Apply ethical theory to current events (politics news stories, etc.)
- Examine the ethical implications of emerging technology

Course Calendar:

Philosophy and ethics are topics that require discussion and debate to demonstrate and increase understanding of the core ideas. This class has weekly readings and active in-class discussion. Much like computer science, the study of philosophy is axiomatic, meaning built up over time; comprehension of earlier readings is just as important to understanding the later readings as the later readings themselves. **Setting sufficient time aside to complete all the readings and/or watch the videos throughout this course is essential to receiving a good grade.** All readings are highly recommended and will likely be referenced in class. They are not “optional” in the traditional sense – I expect you to thoroughly read at least one from each week, hopefully more. However, I do not expect you to read all of them, or the entirety of every selection. Due to their length, you may wish to use AI to summarize them or suggest subsections to peruse if you do not have time to read all the readings in all their entirety. The videos are generally shorter and easier to digest, please don’t skip them. If you find yourself so inclined, please – follow down the citations within the articles and readings, fact-check the videos. Come to class *thoroughly* prepared to discuss the *topic* rather than just the reading.

Week# Date	Material
#0 8/26	Course introduction and syllabus review
#1 9/9	Thinking like a Hacker Selections from <i>Hacker Culture</i> by Douglas Thomas “The Mind Has No Firewall” by Timothy L. Thomas <i>Optional:</i> Selections from <i>Snow Crash</i> by Neal Stevenson
#2 9/16	The Archive and Citation Selection from <i>Algorithms of Oppression</i> by Safiya Noble
#3 9/23	Experts and What They Build <i>Threshold</i> by Ieva Jusionyte (Ch 1, “Tactical Infrastructure”) TEDTalk by Liz Ogbu: https://youtu.be/x0MnGZ1gB4k
#4 9/30	Privacy, Government, and Surveillance (One) Selection from <i>The Rise of Big Data Policing</i> by Andrew Ferguson <i>Optional:</i> Selections from <i>Carceral Capitalism</i> by Jackie Wang

#5 10/7	Privacy, Government, and Surveillance (Two) Selection from <i>This is How They Tell Me the World Ends</i> by Nicole Perlroth Selection from <i>Technically Wrong</i> by Sara Wachter-Boettcher <i>Optional: Watch The Social Dilemma</i>
#6 10/14	Social and Behavioral Hacking (One) Selection from <i>The Facebook Files</i> Selection from <i>Games: Agency as Art</i> by C. Thi Nguyen <u>Final project teams due by midnight 10/16/24</u>
#7 10/21	Social and Behavioral Hacking (Two) Section from <i>A Culture of Conspiracy</i> by Michael Barkun Article by Robert Evans “From Memes to Infowars” <i>Optional: Principia Discordia</i> <u>Reflection 1 due by midnight 10/23/24</u>
#8 10/28	Technological Economy (Automation) Selection from the MIT Task Force Report on Automation. “Technology and Innovation”
#9 11/4	Technological Economy (Attention) Selection from <i>The Age of Surveillance Capitalism</i> by Shoshana Zuboff, “The Reality Business”
#10 11/11	Censorship, Regulation, and Intervention “Some Information is too Dangerous to be on the Internet” by Vincent Muller Selections from <i>Hackers: Heroes of the Computer Revolution</i> by Steven Levy
#11 11/18	No reading – work on final projects in class <u>Reflection 2 due by midnight 11/20/24</u>
#12 12/2	Final Project Presentations

#13	Final Project Presentations
12/9	

Class Policy & Grading:

Participation: 20%

Reflections: 60%

Final Project: 20%

Letter grade cut-offs:

A	93%	C	73%
A-	90%	C-	70%
B+	87%	D+	67%
B	83%	D	63%
B-	80%	D-	60%
C+	77%	F	Below 60%

*These are inclusive boundaries: decimals will **not** be rounded up.*

Attendance Policy:

Philosophy is a complicated subject that builds on a base of knowledge, and learning occurs primarily through discussion and debate. If during this program there is an issue that causes a missed class, please email me so we can plan together.

Academic Conduct:

Most people write how they speak. With the advent of Large Language Models (LLMs) like ChatGPT, text-based deliverables are potentially no longer sufficient to gauge student achievement. As such, you will not get a good grade in this class without reading and trying to understand the material. By the time I receive your final projects, I will have heard you discuss material in front of the class several times. If you try to submit plagiarized work – including the direct output of a Large Language Model – I will likely be able to tell, and I will be forced to follow the University's Academic misconduct policies.

From the student handbook:

Students must adhere to the highest ethical standards as it relates to their academic coursework (i.e., complete their own assignments, quizzes, exams, and any other academic work, etc.). Academic misconduct may result in an "F" for the course, suspension, or expulsion from the program. Students should familiarize themselves with the definition of academic misconduct as it appears in the University of Connecticut Student Code of Conduct booklet:

"Academic misconduct is dishonest or unethical academic behavior that includes, but is not limited to, misrepresenting mastery in an academic area (e.g., cheating), failing to properly credit information, research, or ideas to their rightful originators or representing such information, research, or ideas as your own (e.g., plagiarism)."

Resources:

College is hard, but you do not have to do it alone. There are many resources for those in need of assistance. Please avail yourself of all the services that will help your mental, emotional, and physical well-being. I am willing to listen to your needs, but please note that ***I am not a confidential source of counseling***. I will ***not*** make public anything you tell me, but I may need to bring in others like counselors, deans, or administrators depending on the sensitivity of the info.

Copyright:

My lectures, notes, handouts, and displays are protected by state common law and federal copyright law. Students are authorized to take notes in my class. I will inform you as to whether you are authorized to record my lectures at the beginning of each semester. If you are authorized to record my lectures, you may not copy this recording or any other material, provide copies of either to anyone else, or make commercial use of them without prior permission from me (just ask!).